

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

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**COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2, BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks: 30

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

	A	B	C	D	E	F	G	
Q ₁	2	1.5	2	1	1.5	0.5	1	9.5
Q ₂	2	1	2	1	1.5	0.5	1	9
Q ₃	1.5	1	1.5	1.5	1	0.5	0.5	7.5

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1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

Internet Connection for a Remote Branch Office

The branch office is located in a remote rural area where high-speed wired internet is not available. The network must support cloud applications, video conferencing, and secure communication with the headquarters while maintaining a limited budget.

Recommended Solution

Internet Connection:

A **4G/5G wireless connection or satellite internet connection** can be used as the primary connection. If possible, a second connection can be used as a backup to improve reliability and ensure uninterrupted business operations.

Networking Devices:

- Wireless router or enterprise router
- Managed switch
- Wireless access points
- Backup internet device or secondary connection
- UPS for power backup

Security Mechanisms:

- Firewall to control incoming and outgoing traffic

- VPN for secure communication with headquarters
- Strong passwords and multi-factor authentication
- Regular software and security updates

This solution provides a balance between **cost, reliability, security, and future scalability**. The company can start with the required bandwidth and upgrade the internet plan or add additional networking devices when the number of users increases.

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

Wireless Network Design for a University

The university needs to provide wireless internet access to more than 3,000 students in classrooms, laboratories, libraries, and hostels. The main challenges are supporting many users, reducing signal interference, maintaining security, and working within a limited budget.

Recommended Solution

Access Point Placement:

Access points should be installed in important and high-density areas such as:

- Classrooms
- Laboratories
- Libraries
- Hostels
- Seminar halls

A site survey should be performed before installation to identify areas with weak signal coverage and avoid unnecessary access points.

Frequency Band Selection:

Both **2.4 GHz** and **5 GHz** bands can be used.

- **2.4 GHz** provides wider coverage.
- **5 GHz** provides better speed and reduced interference.

Users should be distributed across different access points and channels to avoid network congestion.

Authentication and Security:

The university can use secure login authentication with:

- Student ID and password
- WPA2/WPA3 wireless security
- Separate network access for students, staff, and guests

This design provides good **coverage, performance, security, and cost efficiency**. Proper access point placement reduces the number of devices required while still providing reliable wireless access to a large number of students.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

Secure Network Architecture for a Financial Institution

A financial institution needs to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The solution must provide strong security without greatly reducing network performance or increasing operational costs.

Recommended Solution

1. Network Segmentation:

The network should be divided into separate segments using **VLANs or subnets**. For example:

- Employee network
- Customer database network
- Online banking servers
- Guest network
- Administrative network

This reduces the risk of unauthorized access and limits the spread of security attacks.

2. Firewall Deployment:

Firewalls should be placed between important network segments and between the internal network and the internet. The firewall can monitor traffic and block unauthorized connections.

3. Authentication:

Strong authentication methods should be used, including:

- Strong passwords
- Multi-factor authentication (MFA)
- Role-based access control

Employees should only be given access to the information required for their job.

4. Intrusion Detection:

An **Intrusion Detection System (IDS)** or **Intrusion Prevention System (IPS)** can be used to monitor network traffic and detect suspicious activities.

This architecture improves **security and compliance** while maintaining good network performance. Using VLANs, firewalls, MFA, and IDS/IPS provides a practical solution that balances **security, performance, and budget**.

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